

COMPUTER SYSTEM VALIDATION PROTOCOL

QMS SaaS Pro

IQ/OQ/PQ Test Plan for ISO 13485:2016 & 21 CFR Part 11 Compliance

This validation protocol establishes the qualification test procedures, acceptance criteria, and documentation requirements for the QMS SaaS Pro system, covering Installation Qualification, Operational Qualification, and Performance Qualification in accordance with GAMP 5 guidelines.

Version: 1.0

Date: 2026-05-03

Classification: Confidential

Table of Contents

1. Purpose and Scope	3
2. System Description	3
2.1 Architecture	3
2.2 QMS Modules	4
2.3 Multi-Tenancy and Security	4
3. Validation Approach	5
3.1 GAMP 5 Categorization	5
3.2 Validation Lifecycle (V-Model)	5
3.3 Test Types and Acceptance Criteria	5
4. Installation Qualification (IQ)	6
4.1 Hardware / Infrastructure Verification	6
4.2 Software Installation Verification	6
4.3 Database Setup Verification	6
4.4 Network and Security Verification	6
4.5 IQ Test Cases	6
5. Operational Qualification (OQ)	7
5.1 Authentication and Authorization	7
5.2 Document Control	8
5.3 CAPA Management	8
5.4 NCR Management	9
5.5 Audit Management	9
5.6 Training Management	9
5.7 Risk Management	10
5.8 Batch Records	10
5.9 Supplier Management	11
5.10 Change Control and Deviations	11
5.11 Dynamic Forms	11
5.12 Compliance Scoring Engine	12

5.13 Multi-Industry Support	12
5.14 Audit Trail (21 CFR Part 11)	12
5.15 Data Export	13
6. Performance Qualification (PQ)	13
6.1 End-to-End Workflow Testing	13
6.2 Concurrent User Testing	14
6.3 Data Integrity Under Load	14
6.4 Disaster Recovery / Backup Verification	14
6.5 PQ Test Cases	14
7. Traceability Matrix	15
7.1 ISO 13485:2016 Traceability	15
7.2 21 CFR Part 11 Traceability	16
8. Deviation Handling	16
9. Final Report and Sign-Off	17
9.1 Validation Summary Template	17
9.2 Sign-Off	17

1. Purpose and Scope

This Validation Protocol defines the formal qualification testing to be performed on the QMS SaaS Pro computer system to demonstrate that it meets its intended requirements and complies with applicable regulatory standards. The protocol establishes the acceptance criteria, test procedures, and documentation requirements for Installation Qualification (IQ), Operational Qualification (OQ), and Performance Qualification (PQ).

The scope of this validation encompasses the complete QMS SaaS Pro application, including all fifteen quality management system modules, the multi-tenant architecture with Supabase Row-Level Security (RLS), dual-mode operation (Demo and Production), electronic signature capabilities, and audit trail functionality. The system boundary extends from the user-facing React/Next.js front-end through the API layer to the Supabase/Prisma persistence layer and all supporting infrastructure components.

This protocol is governed by the following regulatory frameworks and industry standards:

- ISO 13485:2016 — Medical devices, Quality management systems, Requirements for regulatory purposes
- 21 CFR Part 11 — Electronic Records; Electronic Signatures (U.S. FDA)
- EU Annex 11 — Computerized Systems (EudraLex Volume 4)
- GAMP 5 — A Risk-Based Approach to Compliant GxP Computerized Systems
- IEC 62304:2006 — Medical device software, Software life-cycle processes

System Boundary Definition: The validated system includes the QMS SaaS Pro web application (React + TypeScript + Next.js front-end), the Supabase back-end with Prisma ORM and RLS policies, the Zustand state management layer, the 15 QMS functional modules, electronic signature and audit trail subsystems, and the deployment infrastructure. Excluded from this validation are third-party SaaS services consumed via API (e.g., identity providers, email delivery) and the end-user device hardware and browser software.

2. System Description

QMS SaaS Pro is a cloud-native Quality Management System designed for regulated medical device and life sciences companies. It provides a comprehensive suite of fifteen integrated QMS modules that support the full product lifecycle from design control through post-market surveillance. The system is built on a modern web technology stack and operates in a multi-tenant architecture to serve multiple organizations from a single deployment while maintaining strict data isolation.

2.1 Architecture

The system architecture consists of the following principal technology components:

Layer	Technology	Purpose
Front-End	React 18 + TypeScript	Component-based UI with type safety
Framework	Next.js 16 (App Router)	SSR/SSG, API routes, routing

State	Zustand	Client-side state management
Back-End	Supabase + Prisma ORM	PostgreSQL, RLS, auth, storage
Styling	Tailwind CSS + shadcn/ui	Utility-first CSS, UI component library

Table 1: Technology Architecture Stack

2.2 QMS Modules

The system comprises fifteen interconnected quality management modules, each designed to address specific regulatory and operational requirements:

- Document Control — CRUD operations, approval workflows, version control, e-signatures
- CAPA Management — Create, investigate, root cause analysis, effectiveness verification
- NCR Management — Nonconformance reporting, OOS/OOT workflow, disposition tracking
- Audit Management — Planning, execution, findings, CAPA linkage
- Risk Management — FMEA, RPN calculation, risk matrix visualization
- Training Management — Assignment, tracking, completion, overdue alerting
- Batch Records — Manufacturing step execution, QA review, release, lock
- Supplier Management — Qualification, scoring, A/B/C/D rating classification
- Change Control and Deviations — Workflow-driven change management with linkage
- Dynamic Forms — Template builder, form filler, instance locking
- Document Hierarchy — Structured document relationships and navigation
- Compliance Scoring Engine — Per-clause assessment, overall compliance score
- Reports — Data export, management review report generation (CSV, HTML)
- User Management — RBAC, role assignment, session management
- Setup Wizard and Dashboard — Initial configuration, KPI dashboard

2.3 Multi-Tenancy and Security

QMS SaaS Pro employs Supabase Row-Level Security (RLS) policies to enforce tenant isolation at the database level. Every query is automatically scoped to the authenticated tenant, ensuring that no cross-tenant data access is possible. The system operates in a dual-mode architecture: a Demo mode using in-memory storage for training and evaluation, and a Production mode backed by Supabase for live regulated operations. 21 CFR Part 11 electronic signatures are implemented with meaning-based signature semantics (e.g., "Approved by", "Verified by"), timestamp capture, and an immutable audit trail that records all create, read, update, and delete operations across all modules.

3. Validation Approach

The validation of QMS SaaS Pro follows a risk-based approach consistent with GAMP 5 guidelines. This methodology allocates validation effort proportionally to the risk that each system component poses to product quality, data integrity, and patient safety. The approach ensures that critical functions receive thorough testing while minimizing unnecessary effort on low-risk components.

3.1 GAMP 5 Categorization

Category	Description	QMS SaaS Pro Components	Validation Effort
Category 1	Infrastructure Software	Node.js, Bun runtime, Supabase platform, PostgreSQL	IQ verification only
Category 3	Non-Configured Products	Operating system, browser runtime, network stack	IQ verification only
Category 4	Configured Products	Supabase RLS policies, RBAC configuration, notification rules	IQ + OQ
Category 5	Custom Applications	All 15 QMS modules, e-signature engine, audit trail, compliance engine	IQ + OQ + PQ

Table 2: GAMP 5 Category Classification

3.2 Validation Lifecycle (V-Model)

The validation lifecycle follows the V-model, where each development phase has a corresponding verification or validation activity. The left side of the V represents specification and design activities; the right side represents verification and validation activities. This model ensures traceability from user requirements through design specifications to test cases and acceptance criteria.

Specification Phase (Left V)	Verification Phase (Right V)
User Requirements Specification (URS)	User Acceptance Testing (UAT)
Functional Requirements (FRS)	System Integration Testing
Design Specifications (DS)	Module/Component Testing
Code Implementation	Unit Testing

Table 3: V-Model Phase Mapping

3.3 Test Types and Acceptance Criteria

Test Type	Scope	Performed By	Acceptance
Unit Test	Individual functions and components	Development Team	100% pass; coverage >= 80%
Integration Test	Module-to-module interfaces	Development Team	100% pass; no critical defects
Component Test	Complete module functionality	QA Team	100% pass; no major defects

E2E Test	Cross-module workflows	QA Team	100% pass; no major defects
UAT	Business process validation	Business Users + QA	100% pass; no open deviations

Table 4: Test Types, Scope, and Acceptance Criteria

4. Installation Qualification (IQ)

Installation Qualification verifies that the QMS SaaS Pro system and all its components are correctly installed and configured in the target environment according to documented specifications. IQ establishes that the hardware, software, network, and database infrastructure meet the minimum requirements and that all installation steps have been completed successfully.

4.1 Hardware / Infrastructure Verification

Verify that the server infrastructure meets the minimum specifications for CPU, memory, storage, and network connectivity. The production environment shall be provisioned with at minimum 4 vCPU, 8 GB RAM, and 50 GB SSD storage. Network connectivity between the application server and the Supabase instance must be confirmed with latency under 100 ms for standard API calls.

4.2 Software Installation Verification

Confirm that the correct versions of all software dependencies are installed and operational. This includes the Node.js runtime (v20+), Bun package manager, all npm/bun dependencies as specified in the lock file, and the Next.js build artifacts. The build process must complete without errors and the application must start successfully on the designated port.

4.3 Database Setup Verification

Verify that the Supabase instance is provisioned and accessible, the Prisma schema has been migrated successfully, all Row-Level Security (RLS) policies are active and enforce tenant isolation, and initial seed data (reference data, default roles, industry configurations) has been loaded correctly. Database connection parameters must match the environment configuration.

4.4 Network and Security Verification

Confirm that HTTPS/TLS encryption is active for all endpoints, that the Caddy reverse proxy or equivalent gateway is correctly routing traffic, that CORS policies are configured to allow only authorized origins, and that authentication endpoints are accessible. Verify firewall rules restrict database access to the application server only.

4.5 IQ Test Cases

ID	Description	Expected Result	Status
IQ-001	Server hardware meets minimum specs	CPU >= 4 vCPU, RAM >= 8 GB, SSD >= 50 GB	Pass / Fail

IQ-002	Node.js runtime version verified	Node.js v20+ installed and executable	Pass / Fail
IQ-003	Bun package manager installed	Bun v1.0+ installed and executable	Pass / Fail
IQ-004	Application dependencies installed	npm/bun install completes with zero errors	Pass / Fail
IQ-005	Next.js build succeeds	Build completes with no errors; output in .next/	Pass / Fail
IQ-006	Application starts on port 3000	HTTP 200 on / endpoint within 30 seconds	Pass / Fail
IQ-007	Supabase instance accessible	Connection test returns OK; latency < 100 ms	Pass / Fail
IQ-008	Prisma schema migration applied	All tables created; migration log clean	Pass / Fail
IQ-009	RLS policies active	Cross-tenant query returns zero rows	Pass / Fail
IQ-010	Seed data loaded	All reference data tables populated	Pass / Fail
IQ-011	HTTPS/TLS active	Certificate valid; all endpoints HTTPS-only	Pass / Fail
IQ-012	Caddy gateway routing verified	API routes proxied correctly; XTransformPort functional	Pass / Fail
IQ-013	CORS policy configured	Only authorized origins accepted	Pass / Fail
IQ-014	Firewall rules enforced	DB port inaccessible from public network	Pass / Fail
IQ-015	Authentication endpoints functional	Login/logout/signup endpoints return expected codes	Pass / Fail

Table 5: IQ Test Cases

5. Operational Qualification (OQ)

Operational Qualification demonstrates that the QMS SaaS Pro system operates according to its functional specifications throughout all anticipated operating ranges. Each module is tested individually and in combination to verify that functional requirements are met, error conditions are handled gracefully, and data integrity is maintained under all supported operating conditions.

5.1 Authentication and Authorization

The authentication subsystem shall be tested to verify that users can log in with valid credentials, that invalid credentials are rejected with appropriate error messages, that session tokens expire according to policy, and that Role-Based Access Control (RBAC) enforces the defined permission matrix. Each role (Admin, QA Manager, QA Analyst, Production Staff, Viewer) must be verified to access only its authorized functions.

ID	Description	Expected Result	Status
OQ-001	Valid user login	User authenticated; session token issued	Pass / Fail
OQ-002	Invalid password rejection	Error message displayed; no session created	Pass / Fail
OQ-003	Session timeout enforcement	Session invalidated after configured timeout	Pass / Fail

OQ-004	RBAC - Admin full access	Admin can access all modules and functions	Pass / Fail
OQ-005	RBAC - Viewer read-only	Viewer cannot create, edit, or delete records	Pass / Fail
OQ-006	RBAC - Unauthorized access blocked	403 error for unauthorized role/action	Pass / Fail

Table 6: OQ Test Cases - Authentication and Authorization

5.2 Document Control

Document Control is a critical QMS module subject to stringent regulatory requirements. Testing shall verify the complete document lifecycle: creation, review, approval with electronic signatures, publication, version control, obsolescence, and retrieval. The electronic signature workflow must comply with 21 CFR Part 11 requirements for meaning, timestamp, and non-repudiation.

ID	Description	Expected Result	Status
OQ-007	Create new document	Document created with auto-generated ID	Pass / Fail
OQ-008	Submit for review	Document status changes to "In Review"	Pass / Fail
OQ-009	Approve with e-signature	Signature captured with meaning, user, timestamp	Pass / Fail
OQ-010	Reject and return	Document returned to draft with comment	Pass / Fail
OQ-011	Version increment on revision	New version created; prior version archived	Pass / Fail
OQ-012	Obsolete document	Document marked obsolete; removed from active list	Pass / Fail
OQ-013	Document search and retrieval	Search returns correct results by title/ID/type	Pass / Fail

Table 7: OQ Test Cases - Document Control

5.3 CAPA Management

CAPA (Corrective and Preventive Action) Management shall be tested to verify the complete CAPA lifecycle: identification, investigation, root cause analysis, action plan creation, implementation tracking, and effectiveness verification. Linkages between CAPAs and their source records (NCRs, audit findings, complaints) must be maintained and traceable.

ID	Description	Expected Result	Status
OQ-014	Create CAPA record	CAPA created with unique ID; status = Open	Pass / Fail
OQ-015	Link CAPA to source NCR	Bi-directional link established and displayed	Pass / Fail
OQ-016	Record investigation findings	Investigation data saved; status updated	Pass / Fail
OQ-017	Document root cause analysis	Root cause recorded with Ishikawa/5-Why method	Pass / Fail
OQ-018	Create action plan	Actions assigned with owners and due dates	Pass / Fail
OQ-019	Close CAPA after effectiveness check	CAPA closed; effectiveness verified	Pass / Fail
OQ-020	Overdue CAPA alert	Notification triggered when action past due	Pass / Fail

Table 8: OQ Test Cases - CAPA Management

5.4 NCR Management

Nonconformance (NCR) Management shall be tested to verify that nonconformances can be created, categorized, investigated, and dispositioned. The Out-of-Specification (OOS) and Out-of-Trend (OOT) workflow must correctly escalate and route records. Disposition decisions must be recorded with electronic signatures and linked to any resulting CAPA records.

ID	Description	Expected Result	Status
OQ-021	Create NCR record	NCR created with unique ID; status = Open	Pass / Fail
OQ-022	Categorize as OOS/OOT	Category recorded; escalation triggered per rules	Pass / Fail
OQ-023	Record containment actions	Containment documented; status = Contained	Pass / Fail
OQ-024	Link NCR to CAPA	Bi-directional link established	Pass / Fail
OQ-025	Record disposition with e-sig	Disposition saved with signature meaning and timestamp	Pass / Fail
OQ-026	Close NCR	NCR closed; all required fields populated	Pass / Fail

Table 9: OQ Test Cases - NCR Management

5.5 Audit Management

Audit Management shall be tested to verify the complete audit lifecycle: planning, scheduling, execution, findings documentation, and CAPA linkage. Both internal and supplier audits must be supported. The system must track audit timelines and notify stakeholders of upcoming and overdue audits.

ID	Description	Expected Result	Status
OQ-027	Create audit plan	Audit plan created with scope, criteria, and schedule	Pass / Fail
OQ-028	Schedule audit	Audit date assigned; calendar notification sent	Pass / Fail
OQ-029	Record audit findings	Findings classified (Major/Minor/Observation)	Pass / Fail
OQ-030	Link finding to CAPA	CAPA auto-created from finding; link established	Pass / Fail
OQ-031	Close audit	All findings addressed; audit report generated	Pass / Fail
OQ-032	Overdue audit notification	Alert triggered for audits past scheduled date	Pass / Fail

Table 10: OQ Test Cases - Audit Management

5.6 Training Management

Training Management shall verify that training assignments can be created and tracked, that completion is recorded with evidence, and that overdue training triggers escalation alerts. The system must maintain a training matrix linking job roles to required competencies and automatically identify gaps when new SOPs are published.

ID	Description	Expected Result	Status
OQ-033	Assign training to user	Training task created; user notified	Pass / Fail
OQ-034	Complete training with evidence	Completion recorded; evidence attached	Pass / Fail
OQ-035	Overdue training alert	Escalation email sent after due date	Pass / Fail
OQ-036	Training matrix gap analysis	Untrained competencies identified per role	Pass / Fail
OQ-037	Training effectiveness evaluation	Effectiveness score recorded and tracked	Pass / Fail

Table 11: OQ Test Cases - Training Management

5.7 Risk Management

Risk Management shall be tested to verify that FMEA (Failure Mode and Effects Analysis) records can be created, that Risk Priority Numbers (RPN) are calculated correctly from Severity, Occurrence, and Detection ratings, and that the risk matrix correctly classifies risks into appropriate zones (Low, Medium, High, Critical). Risk mitigation actions must be trackable and linked to CAPA when required.

ID	Description	Expected Result	Status
OQ-038	Create FMEA record	FMEA created with failure mode details	Pass / Fail
OQ-039	Calculate RPN	$RPN = \text{Severity} \times \text{Occurrence} \times \text{Detection}$; calculation verified	Pass / Fail
OQ-040	Risk matrix classification	Risk correctly classified per defined thresholds	Pass / Fail
OQ-041	Link risk mitigation to CAPA	CAPA created from risk; bidirectional link	Pass / Fail
OQ-042	Risk re-evaluation after mitigation	Updated RPN calculated and compared	Pass / Fail

Table 12: OQ Test Cases - Risk Management

5.8 Batch Records

Batch Record Management shall verify that manufacturing steps can be defined and executed in sequence, that QA review can approve or reject individual steps, that batch release requires appropriate electronic signatures, and that completed batch records are locked against further modification. Step completion must record the operator identity, timestamp, and any measured values.

ID	Description	Expected Result	Status
OQ-043	Create batch record from template	Batch record instantiated with all steps	Pass / Fail
OQ-044	Execute manufacturing step	Step completed; operator and timestamp recorded	Pass / Fail
OQ-045	QA review of step	QA approval/rejection recorded with e-signature	Pass / Fail
OQ-046	Batch release	Release recorded with e-sig; status = Released	Pass / Fail
OQ-047	Lock completed batch record	Record immutable; edit/delete operations blocked	Pass / Fail

Table 13: OQ Test Cases - Batch Records

5.9 Supplier Management

Supplier Management shall verify that suppliers can be qualified, scored, and classified into A/B/C/D rating categories. Supplier evaluation criteria must be configurable, and the scoring algorithm must correctly compute aggregate scores from individual criteria ratings. Approved supplier lists must be maintained and accessible from procurement workflows.

ID	Description	Expected Result	Status
OQ-048	Create supplier record	Supplier created with all required fields	Pass / Fail
OQ-049	Qualify supplier	Qualification status updated; approval recorded	Pass / Fail
OQ-050	Score supplier evaluation	Aggregate score calculated correctly	Pass / Fail
OQ-051	Assign A/B/C/D rating	Rating auto-assigned per score thresholds	Pass / Fail
OQ-052	Disqualify supplier	Status changed; linked records flagged	Pass / Fail

Table 14: OQ Test Cases - Supplier Management

5.10 Change Control and Deviations

Change Control and Deviations testing shall verify that change requests follow the defined approval workflow, that deviations can be recorded and linked to their source, and that both change controls and deviations support linkage to CAPA, NCR, and risk records. The approval chain must enforce electronic signatures at each decision point.

ID	Description	Expected Result	Status
OQ-053	Initiate change request	Change request created; workflow started	Pass / Fail
OQ-054	Approve change with e-signature	Approval captured; next step enabled	Pass / Fail
OQ-055	Record deviation	Deviation logged with classification and impact	Pass / Fail
OQ-056	Link deviation to CAPA	Bi-directional link established	Pass / Fail
OQ-057	Close change control	All actions verified; change control closed	Pass / Fail

Table 15: OQ Test Cases - Change Control and Deviations

5.11 Dynamic Forms

Dynamic Forms testing shall verify that form templates can be created using the template builder with various field types (text, number, date, dropdown, checkbox, attachment), that form instances can be filled and submitted, and that completed form instances can be locked to prevent further modification. Template versioning must be supported and prior template versions must remain accessible.

ID	Description	Expected Result	Status
OQ-058	Create form template	Template saved with all configured field types	Pass / Fail
OQ-059	Fill form instance	All field types render and accept input correctly	Pass / Fail

OQ-060	Submit form instance	Data saved; status updated to Submitted	Pass / Fail
OQ-061	Lock form instance	Instance immutable after lock; edit blocked	Pass / Fail
OQ-062	Template versioning	New version created; prior version accessible	Pass / Fail

Table 16: OQ Test Cases - Dynamic Forms

5.12 Compliance Scoring Engine

The Compliance Scoring Engine shall be tested to verify that per-clause assessments can be recorded for each applicable regulatory standard, that the overall compliance score is calculated as a weighted aggregate of individual clause assessments, and that the scoring algorithm produces correct results across the full range of input values. Score changes must be reflected in the dashboard within the configured refresh interval.

ID	Description	Expected Result	Status
OQ-063	Record per-clause assessment	Assessment saved with evidence reference	Pass / Fail
OQ-064	Calculate overall compliance score	Score = weighted average; verified by manual calc	Pass / Fail
OQ-065	Dashboard score update	Compliance score refreshes within 30 seconds	Pass / Fail
OQ-066	Score history tracking	Prior scores retained and accessible in history	Pass / Fail

Table 17: OQ Test Cases - Compliance Scoring Engine

5.13 Multi-Industry Support

Multi-Industry Support testing shall verify that the five industry configurations (Medical Devices, Pharmaceuticals, Biotechnology, IVD/In Vitro Diagnostics, and Combination Products) can each be selected and applied during setup, that the correct regulatory checklists are loaded for each industry, and that industry-specific fields and workflows are activated appropriately.

ID	Description	Expected Result	Status
OQ-067	Select Medical Device config	MD-specific checklist and fields loaded	Pass / Fail
OQ-068	Select Pharma config	GxP-specific checklist and fields loaded	Pass / Fail
OQ-069	Select Biotech config	Biotech checklist and fields loaded	Pass / Fail
OQ-070	Select IVD config	IVDR-specific checklist loaded	Pass / Fail
OQ-071	Select Combo Product config	Combined MD/Pharma checklist loaded	Pass / Fail

Table 18: OQ Test Cases - Multi-Industry Support

5.14 Audit Trail (21 CFR Part 11)

The Audit Trail subsystem is a critical 21 CFR Part 11 compliance component. Testing shall verify that all create, read, update, and delete operations are recorded with the user identity, timestamp, previous value (for updates), new value, and the reason for the change. The audit trail must be tamper-evident: no user, including administrators, may modify or delete audit trail entries. The audit trail must support filtering and

export for regulatory inspection.

ID	Description	Expected Result	Status
OQ-072	Record creation logged	Audit entry created with user, timestamp, record ID	Pass / Fail
OQ-073	Record update logged	Previous and new values captured in audit entry	Pass / Fail
OQ-074	Record deletion logged	Deletion entry includes record ID and user	Pass / Fail
OQ-075	Audit trail tamper evidence	Admin cannot modify/delete audit entries	Pass / Fail
OQ-076	Audit trail filtering	Filter by user, date range, module, action type	Pass / Fail
OQ-077	Audit trail export	Export to CSV with all fields intact	Pass / Fail
OQ-078	Reason for change required	Update blocked without reason entry	Pass / Fail

Table 19: OQ Test Cases - Audit Trail (21 CFR Part 11)

5.15 Data Export

Data Export testing shall verify that all supported export formats (CSV and HTML management review report) generate complete and accurate output. Exported data must match the on-screen data exactly, including all fields, formatting, and calculated values. Large data sets must export without timeout or truncation. HTML reports must be formatted for printing and include headers, footers, and page numbers.

ID	Description	Expected Result	Status
OQ-079	CSV export of document list	All columns exported; data matches on-screen display	Pass / Fail
OQ-080	CSV export of CAPA records	All fields including linked records exported	Pass / Fail
OQ-081	HTML management review report	Report includes all sections; formatted for print	Pass / Fail
OQ-082	Large dataset export (>1000 records)	Export completes without timeout; data intact	Pass / Fail
OQ-083	Export date/time stamp	Export file includes generation timestamp	Pass / Fail

Table 20: OQ Test Cases - Data Export

6. Performance Qualification (PQ)

Performance Qualification demonstrates that the QMS SaaS Pro system consistently performs according to its specification under real-world operating conditions. PQ testing validates end-to-end business workflows, concurrent user capacity, data integrity under load, and disaster recovery capabilities. PQ is conducted in the production-equivalent environment with production-representative data volumes.

6.1 End-to-End Workflow Testing

End-to-end workflow testing exercises complete business processes that span multiple QMS modules. Each workflow must complete successfully from initiation to closure, with all cross-module linkages,

notifications, and status transitions verified. Workflows shall be executed by representative end-user roles following documented standard operating procedures.

6.2 Concurrent User Testing

Concurrent user testing verifies that the system maintains acceptable performance and data integrity when multiple users access the system simultaneously. Testing shall simulate at minimum 25 concurrent users performing representative operations across different modules. Response times must remain within defined thresholds: page loads under 3 seconds, API responses under 1 second, and report generation under 30 seconds. No data corruption or cross-tenant data leakage may occur under concurrent load.

6.3 Data Integrity Under Load

Data integrity testing under load shall verify that database transactions are correctly isolated, that concurrent edits to the same record are handled with appropriate optimistic locking or conflict resolution, that audit trail entries are recorded without loss even under high transaction volume, and that RLS policies maintain tenant isolation during concurrent multi-tenant access patterns.

6.4 Disaster Recovery / Backup Verification

Disaster recovery testing shall verify that the Supabase automated backup mechanism creates recoverable snapshots, that a point-in-time recovery can be completed within the defined Recovery Time Objective (RTO) of 4 hours, that the Recovery Point Objective (RPO) of 1 hour is met, and that restored data is complete and consistent with no data loss beyond the RPO window.

6.5 PQ Test Cases

ID	Description	Expected Result	Status
PQ-001	E2E: NCR to CAPA workflow	NCR created, investigated, CAPA raised, closed; full audit trail	Pass / Fail
PQ-002	E2E: Audit planning to closure	Audit planned, executed, findings recorded, CAPA linked, closed	Pass / Fail
PQ-003	E2E: Document lifecycle	Create, review, approve (e-sig), publish, revise, obsolete	Pass / Fail
PQ-004	E2E: Batch record execution	Template to execution, QA review, release, lock	Pass / Fail
PQ-005	E2E: Supplier qualification	Create, evaluate, score, rate, approve supplier	Pass / Fail
PQ-006	25 concurrent users	All users complete operations; no errors or data loss	Pass / Fail
PQ-007	Page load under load	95th percentile page load < 3 seconds	Pass / Fail
PQ-008	API response under load	95th percentile API response < 1 second	Pass / Fail
PQ-009	Concurrent record edit conflict	Optimistic lock prevents overwrite; user notified	Pass / Fail
PQ-010	Audit trail under high volume	All entries recorded; no gaps in sequence	Pass / Fail

PQ-011	RLS isolation under concurrent tenants	Zero cross-tenant data access detected	Pass / Fail
PQ-012	Backup creation and verification	Backup created; checksum validated	Pass / Fail
PQ-013	Point-in-time recovery	Recovery completed within 4-hour RTO; data intact	Pass / Fail
PQ-014	RPO verification	Data loss within 1-hour RPO window	Pass / Fail
PQ-015	Data consistency after recovery	All records match pre-failure state within RPO	Pass / Fail

Table 21: PQ Test Cases

7. Traceability Matrix

The traceability matrix maps regulatory requirements to specific test cases, ensuring complete coverage of ISO 13485:2016 clauses and 21 CFR Part 11 requirements. Each requirement must have at least one test case that verifies compliance. Gaps in traceability must be documented and resolved before the validation is considered complete.

7.1 ISO 13485:2016 Traceability

Requirement	ISO 13485 Clause	Test Case ID(s)	Status
Document Control	Clause 4.2 (Documents)	OQ-007 to OQ-013	Covered
Management Responsibility	Clause 5 (Management)	PQ-001 to PQ-005	Covered
Resource Management	Clause 6 (Resources)	OQ-033 to OQ-037	Covered
Product Realization - Planning	Clause 7.1 (Planning)	OQ-043 to OQ-047	Covered
Design and Development	Clause 7.3 (Design)	OQ-058 to OQ-062	Covered
Purchasing / Supplier Control	Clause 7.4 (Purchasing)	OQ-048 to OQ-052	Covered
Production and Service Provision	Clause 7.5 (Production)	OQ-043 to OQ-047	Covered
Control of Monitoring Equipment	Clause 7.6 (Monitoring)	OQ-063 to OQ-066	Covered
Monitoring and Measurement	Clause 8.2 (Monitoring)	PQ-006 to PQ-011	Covered
Control of Nonconforming Product	Clause 8.3 (NC Product)	OQ-021 to OQ-026	Covered
Corrective Action	Clause 8.5.2 (Corrective)	OQ-014 to OQ-020	Covered
Preventive Action	Clause 8.5.3 (Preventive)	OQ-038 to OQ-042	Covered
Risk Management	Clause 7.1 (Risk)	OQ-038 to OQ-042	Covered
Internal Audit	Clause 8.2.2 (Internal Audit)	OQ-027 to OQ-032	Covered
Change Control	Clause 7.3.7 (Design Changes)	OQ-053 to OQ-057	Covered

Table 22: ISO 13485:2016 Traceability Matrix

7.2 21 CFR Part 11 Traceability

Requirement	21 CFR 11 Clause	Test Case ID(s)	Status
Electronic Signature - Meaning	11.50(a)	OQ-009, OQ-025, OQ-045, OQ-054	Covered
Electronic Signature - Components	11.50(b)	OQ-009, OQ-025, OQ-045, OQ-054	Covered
Signature Manifestation	11.50	OQ-009, OQ-025, OQ-045	Covered
Audit Trail - Creation	11.10(e)	OQ-072 to OQ-074	Covered
Audit Trail - Tamper Evidence	11.10(e)	OQ-075	Covered
Audit Trail - Review and Copy	11.10(e)	OQ-076, OQ-077	Covered
Record Retention and Retrieval	11.10(c)	PQ-012 to PQ-015	Covered
System Access Controls	11.10(d)	OQ-001 to OQ-006	Covered
Authority Checks	11.10(d)	OQ-004 to OQ-006	Covered
Device Checks	11.10(a)	IQ-012 to IQ-014	Covered
Operational System Checks	11.10(f)	PQ-009, PQ-010	Covered
Personnel Qualifications	11.10(i)	OQ-033 to OQ-037	Covered

Table 23: 21 CFR Part 11 Traceability Matrix

8. Deviation Handling

Any deviation from the expected results defined in this protocol must be documented, investigated, and resolved before the validation can be concluded. Deviations are classified into three severity levels based on their potential impact on product quality, data integrity, and patient safety.

Severity	Definition	Action Required
Critical	Direct impact on patient safety or data integrity; system fails to meet a regulatory requirement	Immediate halt of affected testing; root cause analysis; corrective action; retest after fix
Major	Significant functional deficiency that does not directly impact patient safety	Document deviation; impact assessment; corrective action plan; retest within defined timeline
Minor	Cosmetic or documentation issue with no functional impact	Document deviation; assess for cumulative impact; correction in next release

Table 24: Deviation Severity Classification

Deviation Documentation Requirements: Each deviation must be recorded on a Deviation Report form that includes: the test case ID and description, the expected result versus actual result, the severity classification, the root cause analysis findings, the corrective action taken, the retest results, and the sign-off by the Validation Lead and QA Manager. All deviation reports must be appended to the Final Validation

Report as attachments.

Impact Assessment Process: When a deviation is identified, the Validation Lead shall conduct an impact assessment to determine whether the deviation affects other test cases, other modules, or previously completed qualification activities. If the deviation is found to invalidate prior testing results, the affected test cases must be re-executed after the corrective action is implemented and verified.

Resolution and Retesting: Corrective actions must be implemented, verified, and documented before retesting commences. The retest must follow the identical test procedure as the original test case and must achieve a passing result. If the retest fails, the escalation procedure defined in the Quality Manual shall be followed.

9. Final Report and Sign-Off

Upon completion of all IQ, OQ, and PQ testing activities, a Final Validation Report shall be prepared summarizing the validation results. The report must include: a summary of all test cases executed with pass/fail results, a list of all deviations identified and their resolution status, an assessment of compliance with ISO 13485:2016 and 21 CFR Part 11, and a conclusion regarding the system suitability for release to production.

9.1 Validation Summary Template

Qualification Phase	Total Tests	Passed	Failed	Deviations	Result
Installation Qualification	15	___	___	___	___
Operational Qualification	83	___	___	___	___
Performance Qualification	15	___	___	___	___
Overall Total	113	___	___	___	___

Table 25: Validation Summary

9.2 Sign-Off

This validation protocol and its results have been reviewed and approved by the following personnel. By signing below, each signatory confirms that the validation was conducted in accordance with this protocol, that the results are accurately reported, and that the system is recommended for release to the production environment.

Role	Name	Signature	Date
System Owner	_____	_____	___/___/___
QA Manager	_____	_____	___/___/___
Validation Lead	_____	_____	___/___/___

Table 26: Validation Sign-Off